

Robust GLM

1 Exercise 1

In the U.S.A., the food stamp program is a federal government program that was created in order to help low-income people buy food. The U.S. food stamp data were collected in order to study the relation between participation to the food stamp program and various socioeconomic indicators. This data set is well-known in the literature concerning robust estimation and testing in generalized linear models¹.

Here, we have data featuring 150 observations which were randomly selected from the full data set containing information on over 2000 elderly citizens. The response is **Participation**, a dummy indicating the participation to the food stamp program, and the covariates are:

- **Tenancy**, a dummy indicating home ownership,
- **Suppl.Income** indicating whether some form of supplemental income is received,
- and monthly **Income** in U.S. dollars.

There are 24 cases of participation out of the 150 observations in the sample.

- Install and load the **robustbase** library and import the **foodstamp.dat** data. Transform the **Income** variable into $\log(\text{Income}+1)$ and estimate the robust GLM via a quasi-likelihood estimator with tuning constant $c = 1.55$. In the **glmrob()** function, use the "**robCov**" option for the **weights.on.x** argument. Check for potentially influential observations.
- To understand the effect of the tuning constant c , estimate the previous model with $c = 40$. Compare the estimates with those obtained with a classical GLM. What do you observe?
- Back to the robust model with $c = 1.55$: are the variables **Suppl.Income** and **Income** significant? Would we get the same conclusions if we had estimated the model with the classical method ? Explain this using what you found in point **a**).
- Estimate the final model.

¹For more details see Stefanski et al. (1986), Künsch et al. (1989) and Héritier and Ronchetti (1994).